

LightGBM: From First Principles

LightGBM is a fast gradient boosting framework designed to train efficiently on large datasets.

Why LightGBM Is Needed

Boosting can be slow on large data. LightGBM speeds up training using histogram-based splitting and leaf-wise tree growth.

Histogram Splitting

Continuous feature values are grouped into bins. This makes split search faster and reduces memory use.

Leaf-Wise Growth

Many trees grow level by level. LightGBM grows leaf-wise, choosing the leaf that gives the biggest improvement. This can improve accuracy but may overfit if not controlled.

Important Controls

Important controls include number of leaves, max depth, learning rate, number of estimators, subsampling, and regularization.

When to Use It

Use LightGBM for large tabular datasets where speed matters. Be careful with small datasets because leaf-wise growth can overfit.